Highlights

- Multiscale TMM–FvCB–FAO56–TEA–LCA model across three Cameroon perennial crops.
- Penja white pepper yields +7.43% to +8.16% more under semi-transparent OPV.
- Dry-season leaf cooling ($-2.67\,^{\circ}\mathrm{C}$) inverts to monsoon light deficit.
- CW sunlight bounds quantum coherence shift ($|\Delta\eta_{\mathrm{NESS}}| \leq 3.4 \times 10^{-5}$).
- Economic parity vs. c-Si needs R2R OPV $\leq \$15/\mathrm{m}^2$ and $T_{80} \geq 16.3$ yr.